Linux and Vim Commands Lab

Introduction

This lab is designed to help students understand and practice essential Linux commands and Vim editor usage in a VLSI design workflow.

Objectives

- Understand the Linux directory structure and navigate it using basic commands.
- Create, rename, move, and delete files and directories in a Linux environment.
- View, copy, and edit design files from the terminal.
- Manage file permissions and understand ownership using chmod and chown.
- Use the vi editor to Open, create, and edit text files, and navigate through a file.

Essential Linux Commands for VLSI Labs

S. No.	Command	Purpose
1.	pwd	Check current working directory
2.	mkdir	Create directory
3.	cd	Navigate between directories
4.	ls	List directory contents
5.	rm	Remove unwanted files
6.	cp	Copy files
7.	mv	Rename or move files
8.	touch	Create empty files
9.	cat	View the contents of a file on the terminal
10.	clear	Clear the terminal screen
11.	chmod +x	Make files executable
12.	echo	Print text/messages (useful in debugging or scripting)
13.	history	View recently used commands
14.	exit	Close terminal session
15.	find	Locate files
16.	grep	Search for specific text or names in file

Procedure

1. Open Terminal.
2. Use pwd to display current directory.
3. Create a directory: my_vlsi_project
4. Move into it: cd my_vlsi_project
5. Create a file: design.v ,testbench.v
6. View file: cat design.v
7. Rename file: mv sample.txt demo.txt
8. Copy it: cp demo.txt copy_demo.txt
9. Delete a file: rm demo.txt

Instructions to practice the Commands:

- Open terminal
- Create a new directory inside your lab workspace
  `mkdir my_vlsi_project`
- Navigate to the lab directory
  `cd my_vlsi_project`
- Open Vim editor to write your design file and testbench file
  `vim design.v testbench.v`
- Use the ls command to check whether the files were created or not
  `ls`
- Use the cat command to see the contents of your design file and testbench file
  `cat design.v`
- Use the mv command to rename the design.v file to demo.v
  `mv design.v demo.v`
- Use the cp command to copy the contents of demo.v to demo_copy.v
  `cp demo.v demo_copy.v`
- Use the rm command to remove the demo.v file from the directory
  `rm demo.v`

Common Vi commands

Operation Modes While working with vi editor you would come across following two modes :

Command mode : This mode enables you to perform administrative tasks such as saving files, executing commands, moving the cursor, cutting yanking and pasting lines or words, and finding and replacing. In this mode, whatever you type is interpreted as a command.

Insert mode : This mode enables you to insert text into the file. Everything that's typed in this mode is interpreted as input and finally it is put in the file . The vi always starts in command mode. To enter text, you must be in insert mode. To come in insert mode you simply type "i". To get out of insert mode, press the Esc key, which will put you back into command mode.

S No.	Command	Function
1.	i	Enter insert mode.
2.	esc	Return to command mode.
3.	:w	Save the contents of the file.
4.	:q	Exit the vi editor.
5.	:wq	Save the content of the file and then quit the vi editor.
6.	dd	Delete the line in the file.
7.	yy	Copy the line in the file.
8.	p	Paste the line in the file.
9.	u	Undo the changes in the file.
10.	ctrl + r	Redo the changes in the file.
11.	/word	Search for a word in the file.
12.	:s/oldword/newword	Replace the oldword with the newword in the file.

Instructions to practice the Commands:

- Create a file and open it in vim editor: vim notes.txt
- Press `i` to enter Insert Mode.
- Type the following lines:

```
Linux is powerful.
Vi editor Is efficient.
Practice makes perfect.
```

- Press `Esc` to go back to Command Mode.
- Save and exit:

```
:wq
```

- Reopen file, and delete a line using `dd`.
- Undo using `u`.

- Search for “Linux”:

  ```
  /Linux
  ```

- Replace “Linux” with “UNIX”:

  ```
  :%s/Linux/UNIX/g
  ```

Conclusion

In this lab, you learned how to work with essential Linux commands to navigate the file system, manage files and directories, and handle permissions. You also gained hands-on experience with the vim editor, including creating, editing, saving, and modifying text files.
These basic skills are crucial for working efficiently in Linux-based environments, especially in VLSI design and automation workflows.